BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 154 · Paper 1: Networks and Security

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How do you calculate raw text size in bits?

Q2. What can a subprogram's parameter receive?
A. Only integers
B. An array
C. Only strings
D. Only Boolean values

Q3. “readLine() writes a new line into the file.” Correct this. Explain why.

Q4. A program saves a result to a text file. “writeLine(text) only shows text on screen.” Correct this. How should the program save the result?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Network choice — Distinguish LAN/WAN, client-server/peer-to-peer and suitable devices.

Internet services — DNS maps domain names to IP addresses; routers connect networks.

Recognise attacks — Distinguish phishing, malware and denial of service.

Prevent attacks — Combine least privilege, authentication, encryption, updates and backups.

New learning

Network choice: Distinguish LAN/WAN, client-server/peer-to-peer and suitable devices.

Internet services: DNS maps domain names to IP addresses; routers connect networks.

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Prevent attacks: Combine least privilege, authentication, encryption, updates and backups.

Worked example

A school receives a phishing email directing staff to an unfamiliar login page. Recommend two defences.

1. Identify credential theft/social engineering.

2. Train staff to verify suspicious links through another channel.

3. Use MFA so a password alone is insufficient.

4. Explain both measures target different stages of compromise.

We Do · Practise with guidance

Guided 1

A device has poor Wi-Fi but fast wired access. Explain two possible causes.

Worked method: Do not assume internet service itself must be slow.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Correct the comparison so a warning appears on the third failed attempt.

```
failed = 3
if failed > 3 then
    print("Review access")
endif
```

iterate

Question 1

State one role of DNS. [2 marks]

Question 2

Name the device that connects networks. [1 mark]

Question 3

Explain why encryption alone does not prevent phishing. [3 marks]

Exit Ticket · Answer before checking

Encryption stops all social engineering attacks.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Number of characters × bits per character, using the given encoding.

Q2. B — An array

Q3. It reads the next line and advances the current position.

Q4. writeLine(text) writes a line through an open file handle. Open the file, write, then close it. Printing on screen does not save it.

I Do / We Do pages · Guided practice

Q1. Do not assume internet service itself must be slow.

You Do pages · Code First

```
failed = 3
if failed >= 3 then
    print("Review access")
endif
```

Expected result: Review access

You Do pages · Independent questions

Q1. Resolves domain names to IP addresses.

Q2. Router.

Q3. Encryption protects data in transit but cannot stop a user willingly entering credentials into a fraudulent site.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: